Evaluating K-Nearest Neighbors for X-Ray Bone Fracture Detection Using HOG, LBP, and SIFT Feature Extraction Methods

Abyan Farrel Muzakki
Computer Science Department,
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia 11480
abyan.muzakki@binus.ac.id

Simeon Yuda Prasetyo
Computer Science Department,
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia 11480
simeon.prasetyo@binus.ac.id

Eko Setyo Purwanto
Computer Science Department,
School of Computer Science
Bina Nusantara University
Jakarta, Indonesia 11480
eko.purwanto@binus.ac.id

2025 International Conference on Computer Sciences, Engineering, and Technology Innovation (ICoCSETI) | 979-8-3315-0861-6/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICoCSETI63724.2025.11019966

***Abstract*—Musculoskeletal conditions, particularly bone fractures, remain a significant global health issue, with 178 million new cases recorded annually. Accurate and efficient diagnostic tools are critical to assist clinical decisions, particularly in resource-limited settings. This study evaluates the performance of the K-Nearest Neighbors (KNN) algorithm for detecting bone fractures in X-ray images, employing three feature extraction methods: Histogram of Oriented Gradients (HOG), Local Binary Patterns (LBP), and Scale-Invariant Feature Transform (SIFT). A dataset of 10,580 X-ray images, balanced across fractured and non-fractured classes, was used to train and evaluate the models across different K values (1, 3, 5, 7, 9). The results indicate that the HOG-KNN model with K=1 achieved the highest accuracy of 99.03%, outperforming LBP (98.97%) and SIFT (96.79%). These findings underscore the effectiveness of HOG in capturing structural patterns critical for fracture detection. Future research could focus on integrating advanced machine learning techniques, hybrid feature extraction methods, and larger datasets to enhance diagnostic accuracy and applicability in clinical environments.**

Keywords—Bone fracture detection, K-Nearest Neighbors (KNN), Feature Extraction, Histogram of Oriented Gradients (HOG), X-Ray Classification.

I. Introduction

As of 2019, the World Health Organization (WHO) stated there are approximately 1.71 billion people worldwide that have been affected by musculoskeletal conditions (MCs), with osteoarthritis, rheumatoid arthritis, osteoporosis, spinal disorders, low back pain, and severe trauma being the most seen cases among 150 types of MCs [1]. Bone fractures alone contribute to 178 million new fractures globally, with spine, hip, wrist, and upper arm fractures being the most common [2]. Characterized by deterioration found in the muscles, bones, joints, and adjacent connective tissues, MCs could scale up from a constant presence of pain to lifelong limitation in mobility and dexterity. Such obstruction will make it difficult for those affected by hindering their performance to work and participate in society [1]. All of this emphasizes the urgent need for more effective and quick methods that can diagnose MCs.

Along with medical advancements, radiographic imaging for patients affected with MCs has become more accessible. But despite that, slow and missed diagnostics performed by physicians act as a bottleneck in hopes for an effective and fast MC diagnostic method. These misidentified fractures on radiographs account for a considerable portion of diagnostic errors, with interpretation discrepancies sometimes occurring between initial readings by junior physicians and final evaluations by expert radiologists. The total cases of miscalculations and errors represent up to 24% harmful diagnostic discrepancies and are often exacerbated during the evening and night working hours due to fatigue [3].

Similarly, one study also mentions that in the case of chest radiographs (CXRs), despite their accessibility, portability, familiarity, and affordability compared to other imaging tests, CXRs are challenging to interpret and can vary significantly between readers [4]. The need for expertise in order to successfully interpret a radiograph seems illogical since it may not always be available. The average time between image acquisition and radiologist reporting can be as long as 84 hours in some hospital settings, leaving the responsibility of initial image interpretation to emergency department teams. In such circumstances, the diagnostic process often relies on informal readings from junior physicians, whose limited training in radiological interpretation can lead to errors. All of this highlights the need for a consistent and reliable tool that can help physicians (regardless of their experience and other outside factors) correctly identify fractures based on patients' radiographs, particularly in emergency and resource-limited settings [5].

Knowing the problem presented at hand, we can try and utilize artificial intelligence (AI), such as using machine learning (ML) models to create a reliable tool that can help physicians correctly identify fractures based on patients' radiographs. ML itself in recent years has demonstrated substantial potential in addressing the limitations of human radiographic interpretation. These ML models span from traditional ML techniques like K-Nearest Neighbors (KNN) that are easy to implement and understand, up until advanced deep learning (DL) models such as convolutional neural networks (CNNs) [6], [7]. Such methods and models enable computers to "learn" patterns from empirical data, making them able to properly diagnose MCs based on patients' radiographic images [6][8]. Within this particular field, ML has proven itself effective in tasks like fracture detection, often matching or surpassing junior physicians and even expert radiologists in properly diagnosing patients' radiographs [8].

One qualitative review further substantiates the efficacy of ML in fields related to osteoporosis by citing and comparing a total of 89 studies, which are then split up into four broad areas. One of these areas encompasses a total of 32 out of the

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89 studies that compare multiple fracture detection methods using a wide range of ML algorithms. Citing from these 32 studies, this qualitative review reported that the average best accuracy was 89.8% (range 78.4% to 99.1%) and the average best area under the curve (AUC) was 0.92 (range 0.63 to 1.00). Looking deeper into the studies, multiple cases have ML models significantly outperforming human experts, particularly when utilizing advanced techniques like transfer learning and data augmentation. Despite these advancements, this review notes that the success of ML is contingent upon data quality, sufficient sample sizes, and addressing limitations such as cases of overfitting and biased results [9].

Looking back at this previous research and studies, some have yet to elaborate on a thorough analysis of how data preparation methods affect model performance. In particular, nothing is known about how normalization, standardization, and maybe most importantly, feature extraction methods affect the accuracy score of ML models. This study will focus on evaluating the performance of the K-Nearest Neighbors (KNN) algorithm in detecting bone fractures from X-ray images by using three feature extraction methods, such as Histogram of Oriented Gradients (HOG), Local Binary Patterns (LBP), and Scale-Invariant Feature Transform (SIFT). Here the X-ray images are sent through each feature extraction method, and in return, it puts out a simplified and informative version of the original X-ray images. Feature extraction methods are implemented in hopes of increasing the classification accuracy performed by the KNN algorithm. To properly assert the effectiveness of the model, a set of evaluation metrics, including accuracy, precision, recall, F1-score, confusion matrix, and the area under the receiver operating characteristic curve (AUC-ROC), will be used. By establishing a solid framework for examining KNN's ability to increase fracture diagnosis accuracy, these techniques hope to address a critical healthcare issue.

II. Methodology

The proposed experiment is carried through and properly executed by implementing a few methodologies. Starting by implementing the three proposed feature extraction methods and applying the KNN ML models afterwards. Evaluation matrices were also used to ensure effective and convenient evaluation of each feature extraction method that is paired with different iterations of the KNN model.

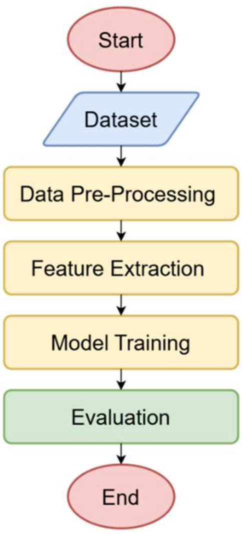

Figure 1. Workflow

Figure 1 represents the workflow and steps that are used in this study, starting from the initial phase where data is accessed from Kaggle [10], followed by the feature extraction methods and the implementation of KNN ML.

A. Dataset

The dataset utilized in this study was sourced from Kaggle, titled "Bone Fracture Multi-Region X-ray Data," published by Madushani Rodrigo [10]. It comprises a total of 10,580 radiographic (X-ray) images, systematically divided into three subsets: training (9,246 images), testing (506 images), and validation (828 images). Each subset is further classified into two main categories: fractured (indicating bone fractures) and not fractured (indicating the absence of fractures). Figure 2 illustrates the distribution of these subsets across the two classes, showing a relatively balanced representation that minimizes potential bias during model training and evaluation.

The dataset includes X-ray images of various anatomical regions, providing a diverse range of visual features necessary for effective feature extraction. Additionally, the images vary in resolution, emphasizing the need for preprocessing to standardize their size and format for feature extraction and machine learning tasks. Figure 3 provides sample images from the training subset, showcasing clear visual distinctions between the fractured and not-fractured classes.

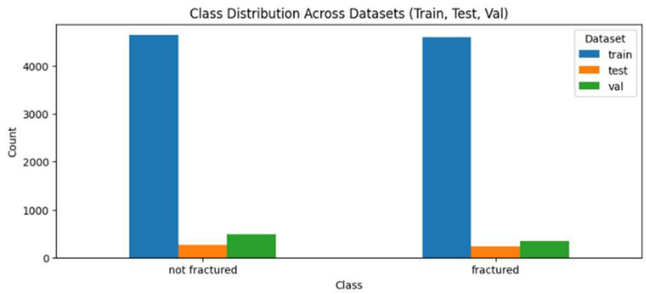

Figure 2. Class Distribution Across Dataset

Figure 3. Sample Images from Train Dataset

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This dataset was chosen for two key reasons. First, its substantial size enables robust model training and reduces the risk of underfitting. Second, the clear annotations for each image allow precise evaluation of the proposed methods. However, a notable limitation is the dataset's narrow focus on fracture detection, potentially limiting the generalizability of findings to other musculoskeletal conditions or imaging modalities. Future work should consider incorporating datasets with broader representations of abnormalities to enhance the applicability of the models.

B. *Data Pre-Processing*

Preparing the data for additional processing in the suggested workflow is the goal of preprocessing photographs. This step improves the image quality and can make the data more consistent so that it works best when other processing or methods are applied [11]. While this study implements a simple form of data preprocessing, this step could be further enhanced to take into account the more complex and imperfect real-world data. Here, the existing dataset that was already split into three files, training, test, and validation, each of which had two subclasses—"fractured" and "not fractured"—results in no further need for data splitting and can move on towards data cleaning. In order to best suit the data before implementing it towards the ML model, it is decided in this study that the data will be sorted in order to decrease the potential errors that might occur during the whole process.

C. *Feature Extraction Methods*

Feature extractions are essentially a way to transform raw data into new, simplified representations while still retaining important information about the original data, with HOG and LBP being the most commonly used feature extraction methods. In this study, three feature extraction methods were used: those being HOG, LBP, and SIFT which are briefly explained below.

Oriented gradient occurrences in certain regions of a picture are counted by the HOG method. The original gradients' distribution serves as a feature vector. The fundamental tenet of HOG is that edge directions, or the distribution of density gradients in an image, may describe the form and appearance of a local item. A histogram of gradient directions is created for each pixel in each of the image's cells, which are tiny, interconnected areas. These histograms are used to form the HOG feature vector that describes a picture [11]. In training and classification, the HOG characteristic is essential. By extracting gradient and orientation, it provides a pixel's shape and direction in the image. The image is divided into smaller areas by the HOG descriptor, and a histogram is produced for each section [12], [13].

To apply the LBP technique, a picture is initially separated into its individual cells in order to extract the features. The associated pixel for every pixel in the cell is compared to every pixel in the 8-neighborhood, where either clockwise or counterclockwise tracing of the pixels and the neighborhood is done. The neighbor pixel's temporary value changes to 0 if the value of the linked pixel is greater than the neighbor pixel's value and to 1 otherwise. Next, an eight-digit number is produced by this procedure, with each produced number's histogram calculated on the cell. A feature vector is then created by merging all of the cell histograms [11], [13], [14].

SIFT algorithms are broken down into four phases. First, features are identified under different image sizes by employing the feature point identification step, which is based on the scale invariance property. Next, subpixel localization refines the placements of these feature points to identify subpixels while eliminating any badly localized key point features that have been detected in the scale space. Last, an orientation histogram is then created using the gradient orientations of sample points inside an area [13], [15]. With this, SIFT's texture descriptor can bring up localized image patterns and is applicable for discriminating regions within an image [16].

D. *Machine Learning Model*

The k-nearest neighbors (KNN) algorithm is a supervised machine learning algorithm predominantly used for classification purposes and could yield good accuracy [10]. By considering the features and labels of the training data, the supervised KNN algorithm predicts the classification of unlabeled data [17]. KNN works by classifying a test sample by attempting to identify its closest neighbors based on the value of k. Using a distance measure, the algorithm first determines the k nearest neighbors before classifying them. The class with a majority is assigned to the test sample class after counting the classes of the closest neighbors. An odd number, k, is selected as the parameter [18], shown in figure 4 below.

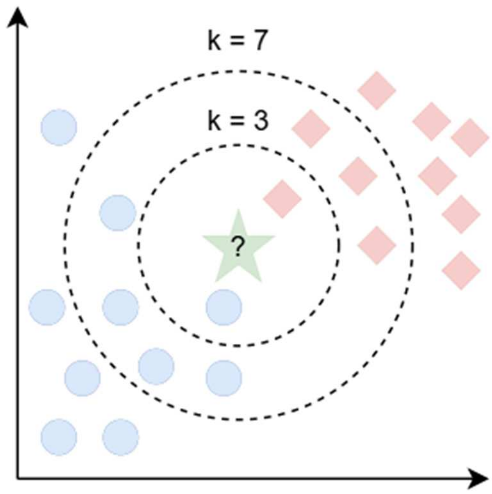

Figure 4. Visualization of KNN Algorithm with two class

Its simplicity in implementation and comprehension is one of its advantages. Additionally, it employs a memory-based methodology and may be rapidly trained using a fresh dataset. However, there is a significant chance that the model will overfit if the K value is not selected properly [7].

E. *Evaluation Metrics*

Four main indicators were used in this study to assess the performance of each method. These include accuracy (Acc), precision, recall, and F1 score. The letters TP, TN, FP, and FN stand for true positive (correctly classified as having a fractured), true negative (correctly classified as having a non-fractured), false positive (erroneously classified as having a fractured), and false negative (erroneously classified as having a non-fractured). All the following metrics used for evaluating the classification performance are given below [18].

Accuracy: Is the ratio between correctly classified bone fractures and the total number of bone fractures [18]. The following is the accuracy equation:

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$$\text{Acc} = \frac{TP + TN}{TP + FP + TN + FN} \quad (1)$$

Precision: Is an indication that shows how much of the positive predictions were actually positive [19]. In this case, the proportion of bone fractures that are correctly identified by the suggested method is represented by Precision (P). The following is the precision equation [11]:

$$Prec = \frac{TP}{TP + FP} * 100 \quad (2)$$

Recall: Also called sensitivity, it refers to the ability of a certain model to correctly identify the presence of pathology [19]. In this case, the proportion of "non fractured" pictures that the system was able to recall is represented by the computed recall (R) [11]. The following is the recall equation:

$$Rec = \frac{TP}{TP + FN} * 100 \quad (3)$$

F1-Score: Sometimes called the dice similarity coefficient (DSE), F1-score is a metric used to describe the similarity between two outputs [19]. In this case, the F1-score is calculated by taking the harmonic mean of recall (sensitivity) and accuracy, but it does not account for actual negatives/true negatives [18]. The following is the F1-Score equation:

$$F1 = \frac{TP}{TP + \frac{1}{2}(FP + FN)} \quad (4)$$

III. RESULTS AND DISCUSSIONS

In this study, Google Colaboratory (Google Colab) was utilized to implement and evaluate three feature extraction methods—HOG, LBP, and SIFT—paired with the KNN machine learning model. The training dataset that consists of 9,246 X-ray images sourced from Kaggle [10], was processed using Google Colab, a cloud-based Jupyter notebook environment. This platform enabled seamless execution of Python code without requiring additional hardware resources, such as CPU or GPU. The default and free Google Colab service consists of an Intel Xeon CPU with 2 vCPUs (virtual CPUs) and approximately 12GB of RAM that can vary depending on the complexity of the program. This study uses the free services that were provided by Google Colab, and such a method can also be applied towards resource-limited settings.

The experiment involved preprocessing the training dataset, followed by applying the HOG, LBP, and SIFT feature extraction methods to transform the raw data into enhanced representations of itself. Here only the necessary features from the data are extracted while the rest are put to the side. Such a procedure makes the data more optimized for classification processes. The extracted feature sets were then input into the KNN model, implemented via scikit-learn's library, and evaluated across different values of K (1, 3, 5, 7, 9).

To assess the performance of each feature extraction method, metrics including accuracy, precision, recall, and F1-score were calculated. Additionally, a confusion matrix and AUC/ROC curve were generated to provide a comprehensive visualization of the model's predictions across both classes. These results facilitate a detailed comparison of the models, highlighting the most effective combination of feature extraction and KNN parameters. The findings are summarized in Table 1, which presents the performance metrics for all tested configurations.

Table 1. Result Table

Feature	K	Acc	Prec	Rec	F1
HOG	1	99.03%	99.67%	98.39%	99.03%
	3	98.70%	98.92%	98.50%	98.71%
	5	98.32%	98.60%	98.07%	98.33%
	7	98.11%	97.87%	98.39%	98.13%
	9	97.73%	97.34%	98.18%	97.76%
LBP	1	98.97%	99.14%	98.82%	98.98%
	3	97.40%	97.42%	97.42%	97.42%
	5	96.00%	95.93%	96.14%	96.03%
	7	95.08%	95.26%	94.96%	95.11%
	9	94.81%	94.47%	95.28%	94.87%
SIFT	1	96.76%	97.18%	96.34%	96.76%
	3	94.21%	94.77%	93.65%	94.21%
	5	93.67%	93.66%	93.76%	93.71%
	7	92.97%	94.44%	91.39%	92.89%
	9	92.05%	93.16%	90.85%	91.99%

The results of the experiment, including accuracy, precision, recall, and F1-score for each feature extraction method (HOG, LBP, SIFT) and KNN iterations (K=1, 3, 5, 7, 9), are presented in Table 1. The table highlights that the HOG feature extraction method combined with K=1 yielded the highest accuracy of 99.03%, outperforming the other combinations. LBP and SIFT followed with maximum accuracies of 98.97% and 96.79%, respectively, both also achieved at K=1. As the value of K increased, a slight decline in performance was observed across all methods, indicating that smaller values of K are more effective for this dataset.

The confusion matrix and AUC/ROC curve for the best-performing model (HOG with K=1) are presented in Figure 5 and Figure 6, respectively.

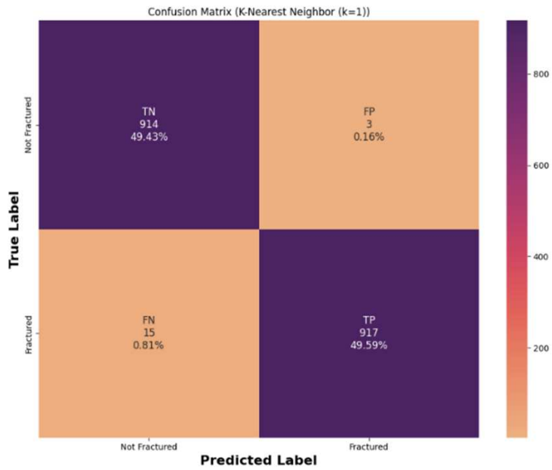

Figure 5. Best Model Confusion Matrix HOG (K=1)

The confusion matrix for the best-performing model (HOG with K=1), shown in Figure 5, provides a detailed breakdown of the model's classification performance. The matrix shows a high number of true positives (correctly classified fractured images) and true negatives (correctly classified not fractured images), with minimal false positives

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and false negatives. This indicates that the model is highly sensitive and specific, ensuring reliable fracture detection without overestimating the presence of fractures.

The near-zero false negative rate is particularly important in medical diagnostics, where failing to detect a fracture can lead to serious consequences. Meanwhile, the low false positive rate minimizes unnecessary interventions, making this model suitable for clinical settings with high patient throughput.

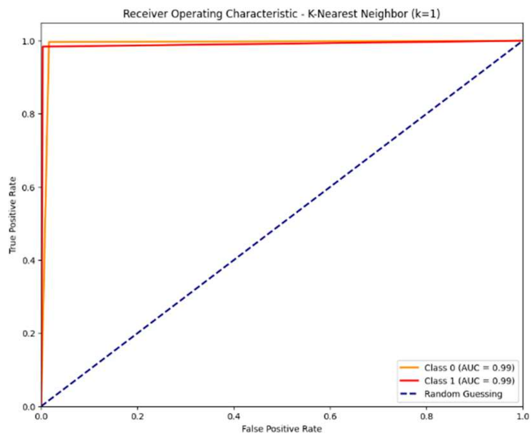

Figure 6. Best Model AUC/ROC Curve HOG-KNN

The AUC/ROC curve for the best-performing model, illustrated in Figure 6, demonstrates its exceptional discriminative ability. The curve rises steeply at the start, indicating a high true positive rate with a minimal increase in false positives. The area under the curve (AUC) approaches 1, confirming that the model can effectively differentiate between the “fractured” and “not fractured” classes under varying decision thresholds.

This strong AUC performance reflects the robustness of the HOG feature extraction method combined with KNN. The curve also validates the model's ability to generalize well across different samples, further supporting its reliability for real-world application.

The experimental findings reveal key insights into the effectiveness of feature extraction methods for bone fracture detection using KNN classifiers. HOG consistently outperformed LBP and SIFT, achieving the highest accuracy of 99.03% at K=1. This superior performance highlights the suitability of gradient-based feature representations for capturing the structural patterns present in X-ray images. While this aligns with expectations for HOG, the results also emphasize the importance of selecting appropriate feature extraction techniques that align with the imaging characteristics of medical data.

LBP and SIFT demonstrated competitive but slightly lower performances, suggesting limitations in their ability to generalize across varying radiographic patterns. The noticeable performance drop at higher K values, particularly for LBP, indicates that the local texture features may introduce noise or fail to effectively scale in higher-dimensional decision boundaries. Similarly, SIFT’s reliance on keypoints may lead to inconsistencies in medical images, which often lack distinctive, high-contrast regions typically suited for keypoint-based descriptors.

A critical takeaway is the importance of feature simplicity and interpretability in medical diagnostics. While methods like HOG provide a compact yet discriminative feature representation, more complex methods such as SIFT may not always align with the domain requirements, particularly when computational efficiency is prioritized.

Another point of interest lies in the robustness of the KNN model. The results indicate that smaller K values (K=1) yield the best performance, but this also raises concerns about overfitting, particularly in cases of noisy or unbalanced datasets. The reliance on KNN’s distance metric further underscores the importance of preprocessing steps like normalization, which were not explored in depth in this study but could have significant implications for real-world deployment.

The dataset used in this study, while diverse and balanced, represents a controlled environment that may not fully capture the variability found in real-world clinical settings. Variations in imaging modalities, noise, and patient demographics can introduce challenges that require more robust and scalable solutions. Future research should aim to address these limitations by employing larger, heterogeneous datasets and incorporating techniques like data augmentation and transfer learning to improve generalizability.

Furthermore, the current study focused solely on KNN as the classification model. While KNN offers simplicity and interpretability, comparisons with more advanced models, such as CNNs or hybrid ensemble techniques, would provide a deeper understanding of the trade-offs between complexity, accuracy, and computational efficiency. Integrating hybrid feature extraction methods or exploring end-to-end learning approaches could also reveal additional avenues for improvement.

IV. Conclusions

This study evaluated the effectiveness of three feature extraction methods—HOG, LBP, and SIFT—paired with a KNN classifier for the detection of bone fractures in X-ray images. The results demonstrated that the HOG-KNN model with K=1 achieved the highest accuracy of 99.03%, outperforming the other combinations. This highlights the suitability of gradient-based feature representations for capturing structural information in radiographic images. The findings further emphasize that feature extraction methods play a critical role in determining model performance, with HOG offering a robust and computationally efficient approach for this task. These results provide a foundation for developing reliable diagnostic tools that can assist in medical imaging analysis, particularly in resource-constrained settings.

Despite the promising results, there are several avenues for future research. First, expanding the dataset to include a broader range of musculoskeletal conditions and more diverse imaging modalities would improve the generalizability of the proposed model. Additionally, such implementation could address the real-world setting where clinical X-ray images might contain noise, artifacts, or variability in image quality. Such an endeavor will require more computational power that can process large amounts of data and complex algorithms. On the other hand, exploring advanced machine learning architectures, such as convolutional neural networks (CNNs) or hybrid models, could offer insights into the trade-offs between interpretability in multiple real-world settings,

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computational cost, and performance. Techniques like data augmentation, domain adaptation, and transfer learning should also be incorporated to enhance model robustness. Finally, the integration of real-time processing capabilities and deployment on edge devices could pave the way for practical applications in clinical settings.

CONTRIBUTORSHIP STATEMENT

Abyan Farrel Muzakki: Contributed to the conceptualization and design of the research project, implemented the HOG, LBP, and SIFT-based feature extraction methods, developed the KNN machine learning model, performed data analysis, and assisted in writing and revising the article.

Simeon Yuda Prasetyo: Provided overall supervision and guidance throughout the research project, curated the dataset, supported coding and debugging during model implementation, enhanced the manuscript through multiple revisions, prepared the final version for submission, and reviewed all sections for accuracy and clarity.

Eko Setyo Purwanto: Supervised the research process and provided critical reviews of the manuscript to ensure its alignment with academic and publication standards.

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